



EXCHEQUER

Smart India Hackathon 2026 · SIH26183 · Ministry of Home Affairs

Exchequer

Research foundations and work to date: the cryptocurrency concepts, the published evidence, and what has been built

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COMPANION TO	The proposal for institutional support of the same date
SOURCES	Every figure is from a document read and quoted in the project's RESEARCH.md

3

blockchains traced

407

verified exchange wallets

234

automated tests

88

real wallets measured

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1 The problem in one paragraph

A victim pays a fraudster in cryptocurrency and reports the address they paid into. Every transaction that address ever made is public, permanent and free to read — but the launderer has moved the money through a dozen throwaway wallets, possibly split it, possibly swapped it for another token, and finally sold it for rupees or dollars at an exchange. The exchange knows who its customer is; the blockchain does not. So the investigator's task is to follow the money across the public part of its journey to the exchange's door, and to arrive there with evidence in a form the exchange's compliance team can match against its own records. Exchequer automates that journey and refuses to guess at any point along it.

2 Cryptocurrency fundamentals

This section is written for a reader who has not needed these concepts before. Each term is defined the way Exchequer uses it; the definitions are conventional.

2.1 The ledger

Blockchain — a public, append-only ledger of transactions, copied across thousands of computers. Once a transaction is recorded in a *block* it cannot be altered or removed, and anyone can read it. This is the single fact that makes tracing possible: the launderer cannot delete the trail.

Transaction — a signed instruction that moves value from one address to another. It carries the sender, the recipient, the amount, the asset, a timestamp (of the block it was included in) and a *transaction hash* — a unique 64-character identifier. The hash is what an officer cites in a legal request.

Address — a public identifier, such as 0x60d02e09... on Ethereum or TQn9Y2khEsLJW1ChVWFMSMeRDow5KcbLSE on Tron, that can receive and send value. An address is controlled by whoever holds its *private key*. Nothing on the blockchain records who that is.

Wallet — software that holds private keys and signs transactions. One person can create thousands of addresses at no cost, which is why launderers use throwaway addresses freely.

Gas — the fee paid to have a transaction included in a block. It explains why amounts shrink by a fraction of a percent at each hop, and why Exchequer's rules carry a 2% tolerance.

2.2 Two kinds of chain, and why we chose one

Account model (Ethereum, BNB Smart Chain, Tron) — each address has a balance, like a bank account, and a transfer reduces one balance and increases another. This makes a trace a walk over a graph of addresses.

UTXO model (Bitcoin) — there are no balances, only unspent outputs of earlier transactions; a payment consumes some and creates new ones, including a *change* output back to the payer. Tracing needs different logic. Exchequer does not cover Bitcoin yet, and says so; the published evidence points at the account-model chains for present-day fraud proceeds (section 4).

2.3 Native coins, tokens and stablecoins

Native coin — the chain's own unit: ETH on Ethereum, BNB on BNB Smart Chain, TRX on Tron. Used to pay gas.

Token — a unit of value defined by a *smart contract* deployed on the chain rather than by the chain itself. On Ethereum and BSC the standard is ERC-20 (BEP-20); on Tron it is TRC-20. A token transfer is recorded as an event emitted by the token's contract, which is why a token trace reads a different data source from a coin trace.

Stablecoin — a token pegged to a currency, almost always the US dollar. USDT (Tether) and USDC (Circle) are the two that matter. They are the launderer's preferred asset because their value does not move while the money is in transit, and because USDT on Tron is cheap and fast to send. Chainalysis measured stablecoins at 84% of illicit cryptocurrency volume in 2025.

2.4 Where the money goes: exchanges and their wallets

Centralised exchange — a company (Binance, Coinbase, CoinDCX, WazirX) that holds customers' cryptocurrency and lets them sell it for rupees or dollars. It is *custodial*: the exchange controls the keys, and by law it knows its customers (KYC). This is the point at which cryptocurrency becomes a bank account, and the only point at which an address becomes a person.

Hot wallet — an exchange-controlled address that holds working funds. Block explorers publish the addresses of the major exchanges' hot wallets under labels such as "Binance 14". Exchequer's attribution is an exact match against those published labels.

Deposit address — the address an exchange hands to one customer to receive their deposit. The exchange controls it, but it is unlabelled, and after a while the exchange *sweeps* the balance into a hot wallet. A deposit address is the address a legal request must name, because a hot wallet receives from thousands of customers and identifies nobody. Exchequer infers deposit addresses from their sweep behaviour (section 6.4).

Cold wallet — an exchange's long-term storage, rarely used; labelled the same way.

2.5 Things that break the trail

Decentralised exchange (DEX) and router — a smart contract that swaps one token for another with no company in the middle (Uniswap, PancakeSwap). A trace that follows ETH ends when the ETH is swapped for USDT, unless the tool reads the transaction's receipt to see what came back. Exchequer does, and

reports where to resume.

Mixer (tumbler) — a service that pools many users' deposits and pays out from the pool, so no payout is linked to any deposit. A trace cannot pass a mixer, and Exchequer stops there by design rather than manufacture a trail. Tornado Cash is the well-known example on Ethereum.

Bridge — a service that moves value to a different blockchain. A bridge to a chain Exchequer does not cover ends the trace.

Explorer and its API — a website (Etherscan, BscScan, TronScan) that indexes the chain so that it can be queried by address. Their free APIs are the data source for the whole project; their rate limits are the reason a first trace takes 20–40 seconds.

2.6 What “tracing” therefore means

Start at the reported address. Read its outgoing transfers. For each recipient, read *its* outgoing transfers, but only those made after the victim's money arrived. Stop when a recipient is a labelled exchange wallet, a sanctioned address, a mixer, or when the depth limit is reached. Along the way, note shapes that launderers use. At the end, say which exchange, which deposit address, which hashes, what amounts, and how confident the arithmetic allows one to be. The picture below is a trace of three hops.



Traced funds in red. Hop 2 is an unlabelled wallet whose only outgoing transfers sweep to Binance 14: a probable deposit address (amber).

3 How laundering looks on a public ledger

Three shapes recur in the literature and in the cases we read. Exchequer detects the first two as fixed rules and reports the third as a stop.

3.1 The peel chain

Named by Meiklejohn et al. in *A Fistful of Bitcoins* (IMC 2013): a large sum sits at one address; a small amount is peeled off to one destination and the remainder passes to a fresh address, and the process repeats, so that the funds walk through a series of single-use wallets and shrink at each step. The paper warns in the same paragraph that the shape “extends well beyond criminal activity” — exchange withdrawals and mining payouts produce it too. Our validation confirms that warning: the rule fires on mining-pool payouts.

3.2 The amount split (fan-out)

One wallet receives a sum and divides it across several recipients, multiplying the paths an investigator must follow. Elliptic's 2025 typology of pig-butcherings proceeds describes funds moved “through dozens of intermediary wallets” before reaching an exchange where “dozens of money mules” convert them. This is sometimes called *structuring*, but that is a legal term (31 CFR § 1010.100(xx)) for splitting *currency* transactions to evade a reporting threshold, and there is no reporting threshold on a public blockchain — so we describe a shape, not an offence. Payroll and exchanges produce the same shape.

3.3 The mixer

Deposits are pooled, payouts come from the pool. Meiklejohn et al. stop their own tracking at mixers for this reason. So does Exchequer. The link does not exist in the data and no amount of traversal recovers it.

3.4 The shape that matters most: the deposit address

Most real cash-outs do not land on a labelled hot wallet. They land on an unlabelled deposit address that the exchange then sweeps. So a trace usually reaches an unlabelled wallet one hop before the exchange — and that wallet is the one to name in a lawful request. The tell is its *whole* history: a deposit address does one thing, every time, to one place. A personal wallet does other things with its money.

4 What the published evidence says, claim by claim

On 15 September 2026 we applied to our own README the standard we apply to attributions: every claim the project makes about laundering was listed, the primary source was fetched and read, the exact passage was copied, and a verdict was recorded. Where the README said more than the source, the README was rewritten. This is the summary; the full document is reproduced as the Appendix, with every source linked.

Claim	Source read	What it says	Verdict
Most illicit crypto moves as stablecoins	Chainalysis 2025 and 2026 Crypto Crime Reports; TRM Labs 2026; FATF June 2025 update (via CNP Law summary)	Stablecoins were 63% of illicit volume in 2024 and 84% in 2025; illicit addresses received at least USD 154 billion in 2025, up 162%, driven mainly by sanctioned entities	Supported, with scope stated
USDT on Tron is the cash-out rail for Indian scam proceeds	UNODC 2024 casino and underground-banking report; TRM Labs 2025; Elliptic 2025; ED case coverage (The Week, July 2026)	UNODC: USDT on Tron is the “preferred choice” of regional cyber-fraud operations. TRM: Tron carried 58% of illicit volume in 2024, halving in 2025. ED froze 160,339 USDT in a Rs. 303 crore case. No source measures India-specific rails	Partially supported; README softened
Peel chains are a laundering pattern	Meiklejohn et al., IMC 2013; Elliptic 2025	Named and described in 2013; still described in 2025. Same paragraph warns it appears in legitimate use. Our thresholds are our own	Supported, with warning
Amount split is a laundering pattern	31 CFR § 1010.100(xx); Elliptic 2025	“Structuring” is a statutory term about currency; on-chain fan-out is a shape analysts describe. Thresholds are our own	Partially supported; wording changed
Laundering hops that matter are close to the source	None	An engineering judgement about where to stop; Meiklejohn followed peel chains for 100 hops	Not supported; kept as a documented cost limit
A mixer's outputs have no link to its deposits	Meiklejohn et al. §6	The paper stops its own tracking at mixers	Supported
Scale of the problem in India	MHA answer to Lok Sabha Unstarred Question 432, 2 Dec 2025	Losses: Rs. 2,290 cr (2022), Rs. 7,465 cr (2023), Rs. 22,845.73 cr (2024); 2.27 million NCRP incidents in 2024	Supported
Issuers and exchanges freeze scam-linked USDT	Tether T3 unit, Jan 2025; Hyderabad police coverage, Jan 2025	USD 100 million frozen globally; Rs. 40 lakh USDT seized in Hyderabad. No Tether announcement names an Indian scam	Partially supported

Two primary sources could not be retrieved (the FATF site returned HTTP 403; the Enforcement Directorate's press pages were unreachable) and are cited through named secondary coverage, labelled as such.

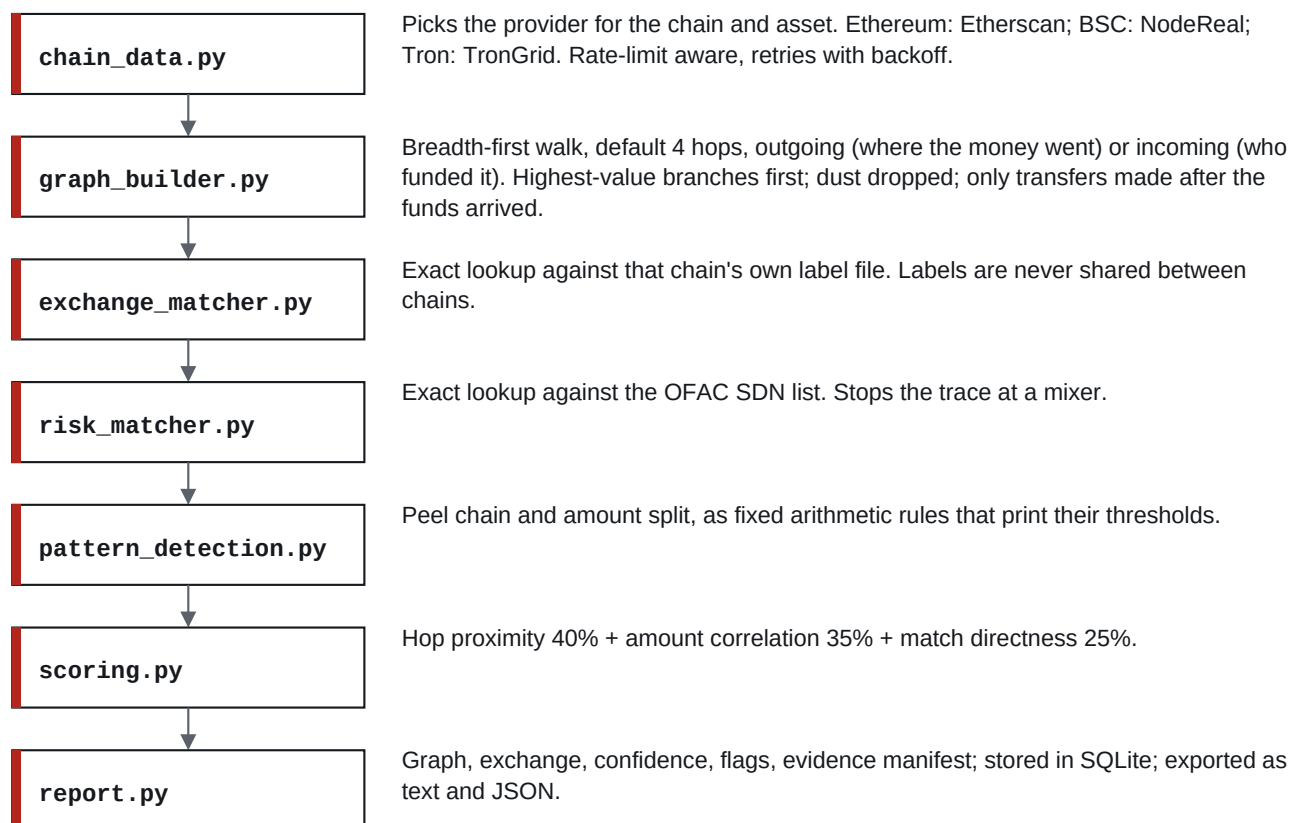
5 The Indian legal framework and what it needs from a tool

- **The offence and the agency.** The Prevention of Money-Laundering Act, 2002 is enforced by the Directorate of Enforcement on the basis of a predicate offence — in the Rs. 303 crore case, two CBI FIRs. A cyber-fraud complaint starts at the state police or the National Cyber Crime Reporting Portal and reaches PMLA through an FIR.
- **Exchanges are reporting entities.** Ministry of Finance notification of 7 March 2023: providers of virtual digital asset services fall under section 2(1)(sa)(vi) of PMLA. FIU-IND's guidelines of 10 March 2023 require them to register.
- **What the exchange must hold.** PMLA section 12(1): a record of all transactions “in such manner as to enable it to reconstruct individual transactions” and documents evidencing the identity of clients and beneficial owners, retained for five years.
- **How an investigator gets it.** PMLA section 50: ED officers may summon any person to produce records. State police obtain the same by notice under section 94 of the Bharatiya Nagarik Suraksha Sanhita, 2023.

What this means for the tool. A trace ends at an address. The record that links that address to a person exists only at the exchange, under section 12, and is obtained under section 50 or a police notice. So every attribution Exchequer prints carries the sentence: *an exchange match identifies where funds arrived, not who controls the account; only the exchange can link a deposit address to a customer identity, via a lawful request.* The report's job is to give the officer the address, the hashes, the amounts and the times, in a form the compliance team can match. That is why the report lists every hash.

6 What we built: the pipeline, stage by stage

Exchequer is a FastAPI backend (Python) and a React frontend, about 9,700 lines of application code, with 234 automated tests and continuous integration. One request to `POST /trace` runs the seven stages below and stores the result as a case.

reported address

6.1 Data: three chains, three providers

Ethereum through Etherscan, BNB Smart Chain through NodeReal, Tron through TronGrid, all on free keys. `chain_data.py` is the only module that knows where data comes from. Each provider is rate-limit aware; we measured Etherscan's free tier and found the documented 5 requests/second is not what is granted — at 0.5 s spacing 4% of requests are refused, at 0.2 s spacing 67% are — so the pacer defaults to the measured optimum. Tron and BSC pagination is bounded by records *read* (1,000), not records kept, after a busy Tron wallet paged its whole history for minutes on the hosted demo.

6.2 The walk, and the time rule

A breadth-first walk to a default depth of 4 hops, following the 10 highest-value counterparties of each address and dropping dust, in either direction. Its one rule about *when*: money cannot be forwarded before it arrives. When the trace reaches a wallet it knows when the traced funds landed, and follows only transfers made at or after that moment. Without this, a busy wallet that once received the victim's funds would have its entire earlier history reported as where the money went. Every node reports how many transfers the rule excluded, so the effect is visible rather than silent. An optional time budget (10–600 s) stops a long trace and reports which addresses were left unexpanded.

6.3 Attribution: exact match, verified labels

Labels are per chain and never shared between chains. 337 Ethereum wallets across 18 exchanges, 30 BSC wallets across 9 (including CoinDCX), 40 Tron wallets across 18 — 407 in total. Ethereum and BSC are imported from a public dataset pinned to one commit; Tron labels are read live from TronScan's own API. Nothing is accepted on the label alone: every address must show no contract bytecode and at least one received transfer on the live chain before it enters the file. Four mislabelled contracts were rejected by that check. WazirX and ZebPay are absent because no source met the standard.

6.4 Deposit-address inference

An unlabelled wallet is a *probable deposit address of exchange E* when it is not the reported address, has made at least two outgoing transfers, and every one of them went to the same labelled E wallet and nowhere else. It is scored as a weaker attribution (match directness 0.5 instead of 1.0), the response says `attribution_inferred: true`, and the report prints the evidence and the sentence that would confirm it. It fired on 0 of 48 control wallets.

6.5 Screening

Every address is checked against the US Treasury's SDN list, generated straight from Treasury's XML with the sanctions programme kept on each entry: 124 Ethereum, 282 Tron and 1 BSC address as of 4 September 2026. A trace stops at a mixer. The designation carries no automatic force in Indian law; it is an escalation trigger.

6.6 Patterns and swaps

Peel chain: runs of at least 2 single-use intermediates, amounts never growing (2% tolerance), ending lower than they started, each hop forwarding at least 50%. Amount split: 4 or more recipients, 90–102% of the inflow passed through, no recipient above 60%, and the address is not itself an exchange. Every finding prints the thresholds it applied. A pattern **never moves the score**; it is a caveat beside it. When a transfer's recipient is a labelled DEX router, the transaction receipt is read and the swap's output asset and amount are reported with where to re-run; the trace is not resumed on the output asset, because amount correlation across 1 ETH and 2,400 USDT has no defensible definition yet.

6.7 The score

Component	Weight	What it measures
Hop proximity	40%	Fewer hops, less room for the trail to be broken by a wallet we cannot see. Linear from 1.0 at a direct deposit.
Amount correlation	35%	How much of the value that left the reported address arrived. Capped at 100%; both figures are printed so dilution is visible.
Match directness	25%	1.0 for a labelled wallet, 0.5 for an inferred deposit address.

No attribution scores *null*, never a low number that might be over-read. There is no model and no clustering, because a conclusion that reaches a courtroom has to be one an officer can re-check by hand.

6.8 Cross-case correlation

Every trace is stored with its full graph, so finding intermediaries shared by traces from *different* complaints is a query, not new tracing. Exchange hot wallets are excluded on purpose (two victims whose money ended at Binance share a bank, not an offender). On 15 September three wallets carrying Etherscan's Phish/Hack label were found to share 61 intermediaries, four of them inferred Binance deposit addresses — the shape of one operation run from several wallets, which three separate complaints would never show.

6.9 Evidence integrity

Every provider response is SHA-256 hashed as it arrives, with the request that produced it (credential removed, parameters in fixed order) and the UTC time, and stored with the case as a manifest. One hash over the sorted records is the manifest hash. The text report prints the manifest as an appendix and ends with a content hash over every byte above it, so the document that leaves the tool can be checked against the document that reaches a court. Cached responses are marked as such with their original retrieval time.

6.10 Interface and hosting

The case view is one finding bar, a hop-layered flow diagram (traced funds in red, zoom and pan), a feed of the transfers that matter, and a folded evidence rail; a bubble graph sits behind a toggle. A present mode

turns a case into four projector screens. A demo sign-in (admin / admin, front-end only, not access control) sits over a live ledger drawn from the stored cases. Reports export as sealed text and JSON. The backend is a Docker image on Railway with a persistent volume; at boot it traces the demo addresses into a disk cache, so the flagship case answers in 0.12 s instead of 43 s.

7 How we know it behaves: validation on real wallets

A threshold that was chosen rather than derived has to be shown to behave. We built two corpora, every address fetched by script from a named source and checked on chain for contract bytecode, with provenance recorded per entry:

- **40 positives** — wallets publicly documented as fraud or theft proceeds: 20 from the OFAC SDN list (Lazarus Group and others), 17 carrying Etherscan's own Phish/Hack label, and the 3 WazirX-hack attacker addresses named by CloudSEK.
- **48 controls** — ordinary wallets with no fraud association from the same pinned dataset the exchange importer uses: funds, mining pools, charities, payment processors, OTC desks, treasuries, airdrop distributors. The pools, processors and distributors are there on purpose: they fan out by design.

Each wallet was traced forward at depth 3 with the real pipeline (1,416 provider requests; 37 addresses per trace on average). 32 positives and 47 controls had at least one outgoing transfer; a rule counts as fired if it produced a finding anywhere in the graph.

Rule	Thresholds	Fired on positives (of 32)	Fired on controls (of 47)
Peel chain	3 transfers, 2 intermediates, 50% retention (defaults)	2	2
Peel chain	3 intermediates	0	0
Amount split	previous defaults: 3 recipients, 50–110%, 90% cap	12	19
Amount split	current defaults: 4 recipients, 90–102%, 60% cap	6	5
Amount split	5 recipients, 90–102%, 60% cap	3	5
Deposit-address inference	2 sweeps, 100% to one labelled wallet	—	0 of 48 seeds

What the numbers establish, and what they do not. **Neither rule separates documented-illicit wallets from ordinary high-volume ones within three hops**, and no point in the 68-point threshold grid brought control firings to zero while still firing on any positive. The amount split at its old defaults flagged 40% of ordinary wallets, which is unusable in a report that names people, so the defaults moved to the grid point with the fewest control firings that still fired on positives — 11% of controls, 19% of positives — an estimate from 79 wallets, not a law. The design already assumed this: a pattern is a reason to look closer, and it never moves the score. What the tool establishes with confidence is different and narrower: that funds reached a wallet published in an exchange label file. 19 of the 32 illicit wallets reached one within three hops; so did 35 of the 47 ordinary ones, because almost everyone's money passes through an exchange. The fraud is established by the complaint; the tool establishes where the money went.

Only Ethereum was measured. No scripted source of Tron controls with the same provenance discipline was found, so no number is claimed for the other chains. The table regenerates offline from the committed snapshot with one command.

8 What it does not do, stated plainly

Every limitation below is also printed in every exported report.

- **One asset per trace.** A swap is detected and reported, not followed; a swap into the native coin, a swap through a liquidity pool, or a bridge to an uncovered chain ends the trail.
- **The graph is a sample.** Only the 10 highest-value counterparties of each address are followed, so a launderer who sends the real money as the eleventh-largest transfer defeats the walk. Raising the limit costs geometrically more requests; it does not remove the gap.
- **Only the newest 200 transfers of each address are read.** On a busy wallet the relevant one may lie beyond.
- **The reported address is not time-windowed,** because the trace does not know when the victim's funds arrived there. Every later hop is.
- **Internal (contract-moved) value is read on Ethereum only,** at double the request cost, and only for contract-shaped addresses on a forward trace. On BSC, Tron and every token trace it is invisible.
- **BSC covers a recent window** (about 17 days by default) because NodeReal caps a query at 100,000 blocks.
- **Attribution is only as good as the label file.** A null result may mean the exchange is absent from it.
- **A trace stops at a mixer and cannot resume.** The link does not exist in the data.
- **Screening covers the OFAC list only;** absence from it establishes nothing.
- **A first-time trace is as fast as the free tier allows:** 20–40 s cold, well under a second cached.

The one thing an evidentiary tool must not do is report money that was never the victim's, and we found ours could — by following transfers made before the funds arrived. That is fixed, tested and written down. We publish the ways a launderer can defeat the tool because its value is in what it *can* establish cheaply and reproducibly, and an investigator has to know where its edge is.

9 Timeline of the work

Date (2026)	What landed
3–5 Sep	Project started as TraceChain: tracing across Ethereum, BSC and Tron; Etherscan pacing measured; first hosted deployment on Railway.
7 Sep	Test suite for attribution and detection rules; reverse tracing; sanctions and mixer screening from the live OFAC list (407 addresses); two verified reverse-trace demos; internal-round presentation.
15 Sep	Adaptive request pacer and response cache; the time rule; every laundering claim sourced and the README softened where sources said less; deposit-address inference; validation on 88 real wallets and the amount-split thresholds tightened; judge Q&A; internal transactions and the service-contract brake; swap detection at DEX routers; cross-case correlation; evidence manifest and content hash; UI direction set (light case-file look).
16 Sep	Renamed Exchequer (a rival team ships as TraceChain); officer sign-in built and then removed at the team's decision (chain of custody rests on evidence hashing instead); persistent cache and warm-up at boot; Railway seed script; GitHub Actions CI.
17 Sep	Optional time budget; hop-layered flow view with zoom, pan and the traced-funds animation; the chequer mark; demo sign-in over the live ledger; present mode; scout script and the procedure for a judge to pick a live address.

Date (2026)	What landed
18 Sep	Tron and BSC pagination bounded by records read after a busy Tron wallet paged for minutes on the hosted demo; tests for both. 234 tests green; main at 4409aea.

65 commits by one author. The internal round used the same judges the next round will.

10 What comes next

- **Before the next round:** no new heuristic rules. Rehearse the four cases (sanctions hit; an attribution the time rule removed, which is the honesty argument; an inferred deposit; three complaints converging). Bring the hosted instance up two days before.
- **Paid tiers:** Etherscan Lite (USD 49/month) unlocks Polygon, Arbitrum, Optimism, Base and Avalanche through code already present; Tron and BSC tiers remove the per-second ceiling.
- **PDF report:** the text report already carries everything; this is a renderer.
- **Bitcoin:** a different model (UTXO); its own adapter.
- **Cross-asset following:** needs a defensible definition of amount correlation across two assets.
- **Engineering debt:** block windowing in the Ethereum client so the 200-newest cap cannot hide a followed transfer; a concurrency guard on /trace; Postgres; frontend tests; streamed progress.
- **A pilot with a cyber-crime cell**, which is the only test that matters.

Against the field

A rival SIH26183 submission we studied has more tests (506), PDF export, a Postgres/Redis worker and six pattern rules — and 17 exchange labels, all Binance on Tron. We have 407 labels across 18 exchanges and three chains, the time rule, cross-case correlation, swap detection, evidence hashing and published failure rates. We would rather be right about less than confident about more.

11 References

All read on 15 September 2026 unless stated; full passages in RESEARCH.md.

1. Chainalysis, *2026 Crypto Crime Report — Introduction*, 8 January 2026; and *2025 Crypto Crime Trends*, 15 January 2025.
2. TRM Labs, *2026 Crypto Crime Report — Key Insights*, 10 January 2026; and *2025 Crypto Crime Report*, p. 6.
3. UNODC, *Casinos, Money Laundering, Underground Banking, and Transnational Organized Crime in East and Southeast Asia*, January 2024, pp. 20, 65.
4. Elliptic, *Typologies Report: Detecting the money flows behind the global pig butchering ecosystem*, 16 October 2025.
5. S. Meiklejohn et al., *A Fistful of Bitcoins: Characterizing Payments Among Men with No Names*, IMC 2013, §5.2 and §6.
6. 31 CFR § 1010.100(xx), definition of structuring (Cornell LII).
7. Ministry of Home Affairs, answer to Lok Sabha Unstarred Question No. 432, 2 December 2025.
8. The Week, *ED uncovers Rs. 303-crore transnational cyber fraud syndicate*, 13 July 2026.
9. Tether, *T3 Financial Crime Unit ... \$100 Million in Criminal Assets Frozen*, 2 January 2025; The Crypto Times, *India Police Seize Rs. 40L in Tether*, 30 January 2025.
10. Prevention of Money-Laundering Act, 2002, sections 12 and 50; Ministry of Finance notification of 7 March 2023; FIU-IND *AML & CFT Guidelines for Reporting Entities Providing Services Related to Virtual*



Digital Assets, 10 March 2023, §5.1; AZB & Partners and Oxford Business Law Blog commentaries (July 2023).

11. CNP Law, *FATF Identifies Stablecoins as a Major Risk*, 11 August 2025 (secondary; FATF site returned HTTP 403).
12. CloudSEK, write-up of the WazirX hack, 19 July 2024 (source of three attacker addresses in the validation corpus).

Appendix. RESEARCH.md, in full

What follows is the project's research file exactly as it stands in the repository on 19 September 2026, converted from Markdown with nothing added or removed. It lists every claim the README makes about laundering, the primary source that was fetched and read for it, the exact passage copied from that document, and a verdict. Where the README said more than the source supports, the README was changed. Every address below is a live link.

Exchequer's pitch is that every attribution can be traced to a published label file. This document applies the same standard to the *claims* the README makes about laundering: each one is listed with the source that supports it, the exact passage, and a verdict. Where the README said more than the evidence supports, the README has been rewritten to say what the sources say.

Rules followed here: every source below was fetched and read during the session of 15 September 2026, and the quoted passages are copied from the document, not paraphrased from memory. Two primary sources could not be retrieved (marked as such); where that happened a secondary source that was read is cited instead, and it is labelled secondary. Nothing is cited that was not read.

Verdicts: **supported** · **partially supported (README overstated)** · **not supported**.

Claim 1 — "Most laundering moves as stablecoins rather than native currency"

Source A. Chainalysis, *2026 Crypto Crime Report — Introduction*, 8 January 2026.

<https://www.chainalysis.com/blog/2026-crypto-crime-report-introduction/>

"stablecoins have come to dominate the landscape of illicit transactions, and now account for 84% of all illicit transaction volume"

"illicit cryptocurrency addresses received at least \$154 billion in 2025 ... This represents a 162% increase year-over-year (YoY), primarily driven by a dramatic 694% increase in the value received by sanctioned entities"

Source B. Chainalysis, *2025 Crypto Crime Trends*, 15 January 2025.

<https://www.chainalysis.com/blog/2025-crypto-crime-report-introduction/>

"stablecoins now occupying the majority of all illicit transaction volume (63% of all illicit transactions)"

Source C. TRM Labs, *2026 Crypto Crime Report — Key Insights*, 10 January 2026.

<https://www.trmlabs.com/resources/blog/2026-crypto-crime-report-key-insights-trm-identifies-record-usd-158-billion-in-illicit-crypto-flows-in-2025-reversing-a-multi-year-decline>

"Inflows to sanctioned entities were predominantly using stablecoins." "Stablecoins served as the primary layer for value transfer across sanctioned jurisdictions, underground banking networks, and illicit service providers." "Top stablecoins included USDT and Russia's ruble-pegged A7A5."

Source D (secondary — the FATF page itself returned HTTP 403 on every attempt). CNP Law, *FATF Identifies Stablecoins as a Major Risk in Illicit Use of Virtual Assets*, 11 August 2025, summarising FATF's sixth Targeted Update on Virtual Assets and VASPs of 26 June 2025.

<https://www.cnplaw.com/fatf-identifies-stablecoins-as-a-major-risk-in-illicit-use-of-virtual-assets/>

"The FATF noted that this trend has continued to grow since the 2024 Targeted Update, and that most on-chain illicit activity now involves stablecoins."

Verdict: supported, with one honest caveat the README now carries. The figures are for *illicit transaction volume* as measured by two analytics firms, and the 2025 jump is driven by sanctioned entities, not by consumer fraud. The README sentence was rewritten to quote the figures and their scope rather than to assert "most laundering".

Claim 2 — "USDT-TRC20 on Tron is the dominant cash-out rail for scam proceeds out of India"

Source E. UNODC, *Casinos, Money Laundering, Underground Banking, and Transnational Organized Crime in East and Southeast Asia: A Hidden and Accelerating Threat*, January 2024, pp. 20 and 65.

https://www.unodc.org/roseap/uploads/documents/Publications/2024/Casino_Underground_Banking_Report_2024.pdf

p. 65: "USDT on the TRON blockchain has become a preferred choice for regional cyberfraud operations and money launderers alike due to its stability and the ease, anonymity, and low fees of its Transactions."

p. 20, note 26: "As of June 2023, TRON had over 165.5 million total user accounts, more than 5.81 billion total transactions, and over US \$11.79 billion in total value locked (TVL), hosting the largest circulating supply of Tether (USDT) globally since April 2021."

Source F. TRM Labs, *2025 Crypto Crime Report* (covering 2024), p. 6.

https://cdn.prod.website-files.com/6082dc5b670562507b3587b4/67f7132b33d3535ca28a54a4_TRM_2025%20Crypto%20Crime%20Report.pdf

"In 2024, the largest percentage of illicit crypto activity occurred on the TRON blockchain (58% of illicit volume), followed by Ethereum (24% of illicit volume), Bitcoin (12% of illicit volume), Binance Smart Chain (3% of illicit volume), and Polygon (3% of illicit volume), reflecting continued preference for blockchains that have low transaction fees, smart contracts, and popular stablecoins."

"However, of all the blockchains analyzed, TRON saw the most significant decline in illicit volume, dropping by USD 6 billion and halving its proportion of illicit volume."

Source G. Elliptic, *Typologies Report: Detecting the money flows behind the global pig butchering ecosystem*, 16 October 2025.

<https://www.elliptic.co/blog/elliptics-typologies-report-detecting-the-money-flows-behind-the-global-pig-butchering-ecosystem>

"At this stage, the money launderers will send the funds to a VASP, where as many as dozens of money mules may be employed to help convert the funds into fiat currencies."

Source H (press coverage of an ED case; no ED press release could be located on the ED website). The Week, *ED uncovers Rs. 303-crore transnational cyber fraud syndicate; 10 arrested, crypto trail traced to Dubai*, 13 July 2026.

<https://www.theweek.in/news/india/2026/07/13/ed-uncovers-indian-rs-303-crore-transnational-cyber-fraud-syndicate-10-arrested-crypto-trail-traced-to-dubai.html>

"converted into cryptocurrency using exchanges such as Binance, GetBit and Carretx. The cryptocurrency was then transferred to private digital wallets allegedly controlled by foreign beneficiaries."

"freezing of around 1,60,339 USDT, 118.73 SOL and 563 ETH"

"The probe was initiated under the Prevention of Money Laundering Act (PMLA) in March 2024 on the basis of two CBI FIRs relating to large-scale cyber fraud."

Verdict: partially supported (README overstated). What the sources establish: USDT on Tron is the rail the UN describes as the "preferred choice" of the Southeast-Asian cyber-fraud industry; Tron carried the majority of measured illicit volume in 2024 and that share halved in 2025; an Indian ED case against a Rs. 303 crore cyber-fraud syndicate ended in USDT being frozen. What no source establishes: a measurement

of *India-specific* cash-out rails, or the word "dominant" for India. The ED coverage names USDT but not the blockchain. The README sentence was rewritten to say exactly this. The `tron_client.py` and `config.py` docstrings, which repeated the claim, were rewritten the same way.

Claim 3 — "Peel chains are a known laundering pattern"

Source I. S. Meiklejohn, M. Pomarole, G. Jordan, K. Levchenko, D. McCoy, G. M. Voelker, S. Savage, A *Fistful of Bitcoins: Characterizing Payments Among Men with No Names*, Proceedings of the ACM Internet Measurement Conference (IMC'13), Barcelona, 23–25 October 2013, §5.2 and §6.

<https://cseweb.ucsd.edu/~smeiklejohn/files/imc13.pdf>

"we focus on an idiom of use that we call a 'peeling chain.' The usage of this pattern extends well beyond criminal activity, and is seen (for example) in the withdrawals for many banks and exchanges, as well as in the payouts for some of the larger mining pools. In a peeling chain, a single address begins with a relatively large amount of bitcoins ... A smaller amount is then 'peeled' off this larger amount, creating a transaction in which a small amount is transferred to one address ... and the remainder is transferred to a one-time change address. This process is repeated"

On tracing a theft through peeling chains: "54 out of 300 peels went to exchanges alone ... the evidence that the deposited bitcoins were the direct result of either a Ponzi scheme or the sale of drugs might motivate Mt. Gox or any exchange (e.g., in response to a subpoena) to reveal the account owner corresponding to the deposit address in the peel"

Source G again (Elliptic, 2025): laundered pig-butcherer proceeds "move the funds through dozens of intermediary wallets - a process known as [chain-peeling]".

Verdict: supported — with a warning the README now states. The pattern and its name come from the 2013 Bitcoin de-anonymisation literature, and Elliptic still describes it in 2025 as a step in laundering scam proceeds. But the paper that coined the term says in the same paragraph that the shape "extends well beyond criminal activity" and is seen in ordinary exchange withdrawals and mining payouts. Exchequer's rule adapts the idea to an account-based chain (there are no change addresses on Ethereum or Tron, so "one-time change address" becomes "single-use intermediate wallet"). **The numeric thresholds — at most 3 transfers, at least 2 intermediates, at least 50% forwarded per hop, 2% growth tolerance — are this project's own choices and come from no paper.** Their behaviour on real wallets is measured in the README section *Validation against real wallets*.

Claim 4 — "Amount split (structuring) is a known laundering pattern"

Source J. 31 CFR § 1010.100(xx), U.S. Treasury / FinCEN definitions, read at

<https://www.law.cornell.edu/cfr/text/31/1010.100>

"a person structures a transaction if that person, acting alone, or in conjunction with, or on behalf of, other persons, conducts or attempts to conduct one or more transactions in currency, in any amount, at one or more financial institutions, on one or more days, in any manner, for the purpose of evading the reporting requirements ... 'In any manner' includes, but is not limited to, the breaking down of a single sum of currency exceeding \$10,000 into smaller sums"

Source G again (Elliptic, 2025): funds are moved "through dozens of intermediary wallets" before reaching a VASP where "dozens of money mules may be employed".

Verdict: partially supported (README overstated). "Structuring" is a defined legal term about *currency* transactions broken up to evade a *reporting threshold*. There is no reporting threshold on a public blockchain, so the on-chain version — one wallet fanning a received sum out across several recipients — is a laundering *shape* described by analytics firms, not structuring in the statutory sense. The README and the

pattern_detection.py docstring no longer say the split is done "to stay under reporting thresholds"; they say what is observable: the sum was divided to multiply the paths an investigator must follow, and the same shape is produced by payroll, exchanges and market makers. **The thresholds — 3 or more recipients, 50–110% forwarded, no single recipient above 90% — are this project's own choices.**

Claim 5 — "Laundering hops that matter happen close to the source; past ~4 hops the graph is mostly unrelated exchange traffic" (graph_builder.py)

No source. This is an engineering judgement about where to stop a breadth-first walk, and the docstring now says so. The only related evidence read is Meiklejohn et al. following peeling chains for 100 hops (Source I), which shows the opposite can be true on Bitcoin. The depth limit is a cost control that the README's Limitations section describes as such.

Verdict: not supported as a claim about laundering; retained as a documented cost limit.

Claim 6 — "A mixer pays out from a commingled pool, so its outputs have no established link to a deposit"

Source I (Meiklejohn et al., §6) describes mixing services as pooling inputs so that following the change link no longer follows one user's money; this is the reason the paper's own peeling-chain tracking stops at them.

Verdict: supported at the level the README uses it (a reason to stop the walk, not a quantitative claim).

Claim 7 — Scale of the problem in India (DEMO.md and JUDGE_QA.md)

Source K (primary). Government of India, Ministry of Home Affairs, Lok Sabha Unstarred Question No. 432, answered 2 December 2025 by the Minister of State for Home Affairs (Shri Bandi Sanjay Kumar).

<https://www.mha.gov.in/MHA1/Par2017/pdfs/par2025-pdfs/LS02122025/432.pdf>

"As per NCRP & CFCFRMS operated by I4C, total amount of losses incurred by citizens due to cyber frauds in the entire country including Bihar from the year 2022 to 2024 are as under: 2022 — 2290.24; 2023 — 7465.18; 2024 — 22845.73 (Rs. in crore)"

Incidents registered on the NCRP: "2022 1029026 ... 2023 1596493 ... 2024 2268346"

Verdict: supported (these figures are new to the project and are quoted exactly).

Claim 8 — "Tether / exchanges freeze addresses tied to scams" and cooperation with law enforcement

Source L. Tether, *T3 Financial Crime Unit Marks Enforcement Victory: \$100 Million in Criminal Assets Frozen Across Five Continents*, 2 January 2025.

<https://tether.io/news/t3-financial-crime-unit-marks-enforcement-victory-100-million-in-criminal-assets-frozen-across-five-continents/>

"frozen more than \$100 million in criminal assets globally" ... cases involving "money laundering, investment fraud, blackmail operations, terrorism financing, and other serious financial crimes" ... "Launched in August 2024".

Source M (press coverage, secondary). The Crypto Times, *India Police Seize Rs. 40L in Tether from Nationwide Fraud*, 30 January 2025, citing the Hyderabad City Police on X.

<https://www.cryptotimes.io/2025/01/30/india-police-seize-%E2%82%B940l-in-tether-from-nationwide-fraud/>

"The Hyderabad Police have successfully uncovered a nationwide cyber fraud ring" ... "the police have also seized the Rs. 40 lakh worth of Tether".

Verdict: partially supported. USDT tied to fraud is frozen at scale by the issuer on request of law enforcement, and Indian police have seized USDT in cyber-fraud cases. **No Tether announcement naming an Indian scam specifically could be found**, so the project makes no such claim.

What the Indian legal framework needs from a tool like this

This section grounds the report's "what would confirm this" language in the process an investigator actually follows.

The offence and the agency. The Prevention of Money-Laundering Act, 2002 (PMLA) is enforced by the Directorate of Enforcement (ED). An ED investigation is opened on the basis of a predicate offence — in the Rs. 303 crore case above, "two CBI FIRs relating to large-scale cyber fraud" (Source H). A cyber-fraud complaint therefore starts at the state police or the National Cyber Crime Reporting Portal (Source K), and reaches the PMLA route through an FIR.

Exchanges are reporting entities. By a Ministry of Finance notification dated 7 March 2023, entities providing services in virtual digital assets (exchange between VDAs and fiat, exchange between VDAs, transfer, custody, and financial services around issuance) were designated as persons carrying on a designated business under section 2(1)(sa)(vi) of the PMLA, which makes them *reporting entities*. Sources read: AZB & Partners, *The Indian Anti-Money Laundering Regime: New Compliance Obligations Around Virtual Digital Assets*

<https://www.azbpartners.com/bank/the-indian-anti-money-laundering-regime-new-compliance-obligations-around-virtual-digital-assets-2/>

Oxford Business Law Blog, *Digital Assets & the Indian Anti-Money Laundering Regime*, July 2023

<https://blogs.law.ox.ac.uk/oblb/blog-post/2023/07/digital-assets-indian-anti-money-laundering-regime>

"the Central Government, via a notification dated March 07, 2023 ('Notification'), has brought activities involving or relating to virtual digital assets ('VDA'), including cryptocurrencies and NFTs under the purview of the PMLA."

Registration and the guidelines. FIU-IND, *AML & CFT Guidelines for Reporting Entities Providing Services Related to Virtual Digital Assets*, effective 10 March 2023, §5.1

https://fiuindia.gov.in/pdfs/AML_legislation/AMLCFTguidelines10032023.pdf

"In terms of Rule 2 (wa) read with Rule 2 (sa) of PMLA, all SPs are required to register as Reporting Entities with FIU-IND."

What the exchange must hold. PMLA section 12(1), read at

<https://indiankanoon.org/doc/290623/>

every reporting entity shall

"(a) maintain a record of all transactions ... in such manner as to enable it to reconstruct individual transactions" ... "(e) maintain record of documents evidencing identity of its clients and beneficial owners as well as account files and business correspondence relating to its clients"

with a retention period of five years (FIU-IND guidelines; AZB summary).

How an investigator gets it. PMLA section 50, read at

<https://indiankanoon.org/doc/1395568/>

(2) "The Director, Additional Director, Joint Director, Deputy Director or Assistant Director shall have power to summon any person whose attendance he considers necessary" to give evidence or produce records; (3) "All the persons so summoned shall be bound to attend ... and shall be bound to state the truth".

Outside the PMLA route, state police obtain the same records from a registered exchange by notice under the criminal procedure code (section 94 of the Bharatiya Nagarik Suraksha Sanhita, 2023, formerly section 91 CrPC) — this step is general procedure and is stated here from the investigator's side, not from a source read in this session.

What this means for Exchequer's output. A trace ends at an *address*. The record that links that address to a person exists only at the exchange, under section 12, and is obtained under section 50 (ED) or a police notice. So the sentence the report attaches to every attribution — "an exchange match identifies where funds arrived, not who controls the account; only the exchange can link a deposit address to a customer identity, via a lawful request" — is precisely what the framework provides for. The report's job is to give the officer the address, the transaction hashes, the amounts and the times, in a form the exchange's compliance team can match against its own records. That is why Phase 8 of this work makes the report list every hash.

For an *inferred* deposit address (Phase 2), the same request confirms or refutes the inference: the exchange either has a customer record for that address or it does not.

Sources that could not be retrieved

- FATF, *Targeted Update on Implementation of the FATF Standards on Virtual Assets and VASPs*, June 2025, and the FATF *Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs*, October 2021 — the FATF website returned HTTP 403 to every fetch (HTML page and PDF). A law-firm summary was read instead and is labelled secondary above.
- Enforcement Directorate press releases — the ED website's press-release pages could not be reached in the session, so ED cases are cited through named press coverage.
- Lexology's summary of the FATF update — HTTP 403.



Exchequer is named for the medieval treasury that counted the Crown's money on a chequered cloth. One square of the mark is red: the sum being traced.